

Solutions to Assignment 4

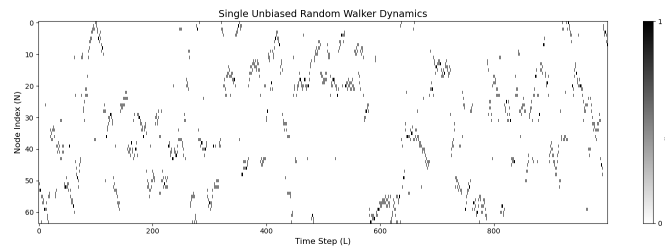
Advanced Tools for Complex Network Analysis

CNET 5052, Spring 2026

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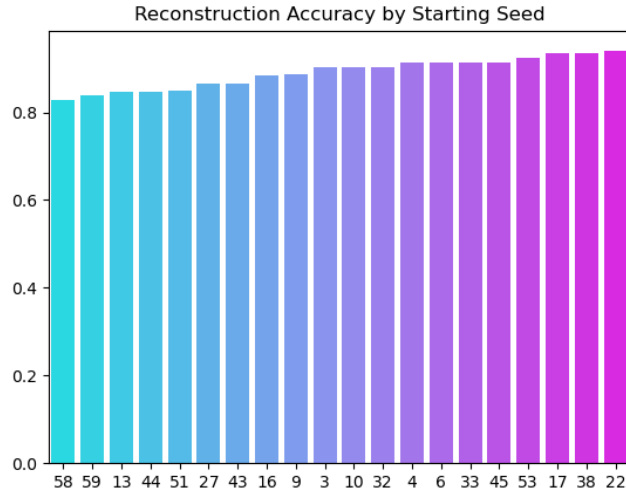
1. Question 1.

- (a) I chose to reconstruct a small-world graph using a random walker as the dynamic. I visualized the first generated time-series matrix using imshow.

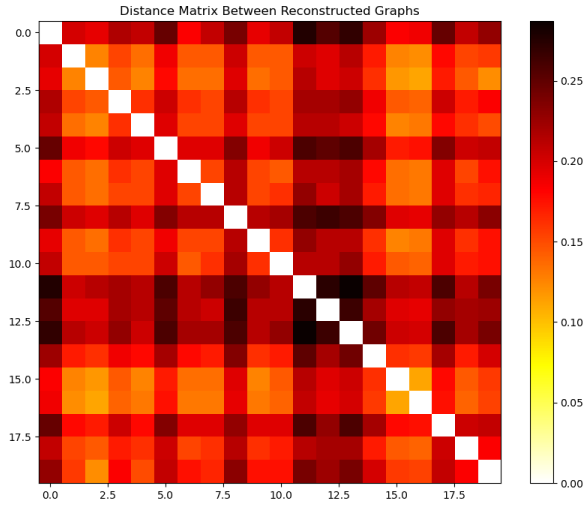


- (b) I used Granger Causality, which is designed to identify causal relationships in time-series by determining if X is useful in forecasting Y to a statistically significant degree compared to using past values of X alone. This is a good choice for single-walker random walk dynamics because these do not involve covariance of nodes (one state at a time).
- (c) See notebook.
- (d) The method that first stands out to me as a good fit for this task is Adjusted Mutual Information, a variant of the Mutual Information score, which works by measuring the extent to which knowing information about an edge in G/R reduces uncertainty about its position in R/G, then adjusting for the bias induced by sparsity (vast majority in "0" class) by measuring the difference between the observed MI score and the expected score under a regime with the exact same number and size of clusters but random assignments.

* ADD CLASS REPO FORK https://github.com/caleb-chandler/cnet5052_sp26



(e) Shown below.



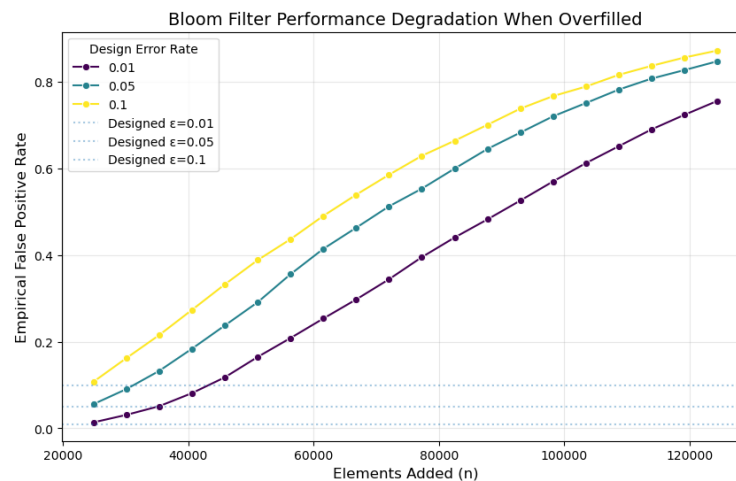
(f) Notwithstanding the somewhat misleading scale discrepancy between the barplot and heatmap, the differences between reconstructed/original and reconstructed/reconstructed are very similar. For low-entropy starting graphs such as this one, this makes logical sense: the reconstructions are all reconstructing the same thing using the same information and the same limited pool of combinations to choose from, so you would expect them to converge on similar results. You would also expect graphs that are different from each other directly to also be different – at about the same magnitude – in terms of their level of difference with the original graph, and indeed the barplot demonstrates exactly this, even if only visually.

But there's no reason this *has* to be the case. If the number of potential ways to reach high similarity is itself high, the reconstructions might actually end up

being quite far from each other, even if their distances from the original graph remain low. I would expect potential distance to scale superlinearly with both the size of the graph and the proximity of density to 0.5, since these would present exponentially more opportunities for high-similarity permutations. Anything in the pipeline – whether dynamics, algorithm, or thresholding approach – that caused either of these things to occur would also cause the average R-to-R distance to increase. And unless my thinking is wrong, the maximal such distance should be hard-capped by the number of possible high-similarity matrix permutations above some given threshold.

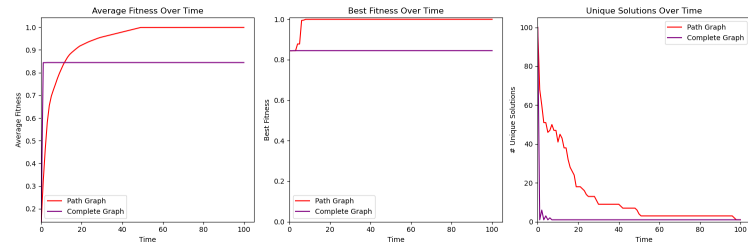
2. Question 2.

- (a) See notebook.
- (b) Checking for existence in the full graph takes $s = \frac{N(N-1)}{2} = 499,500$ seconds; one for each possible edge. The bloom filter instantly reduces the number of such possible edges down to a fraction of that; specifically, the full set of edges that actually exist $e = 0.05s = 24,975$, plus some number of false positives f determined by $\epsilon(s - e)$. The number of required lookups – and thereby the number of seconds – then becomes $e + f$. For $\epsilon = 0.01, 0.05$, and 0.1 , this comes out to $\approx 29,720, 48,701$, and $72,428$ respectively.
- (c) A bloom filter can never lead to a *worse* lookup time, because it can only reduce the size of the set to be checked, never increase it. You will, however, start getting significantly less benefit out of it the more incongruent your array size becomes with the number of elements added to it, as your designed error rate shrinks further into irrelevancy as the actual error rate balloons way out of scope. The following chart demonstrates this issue.



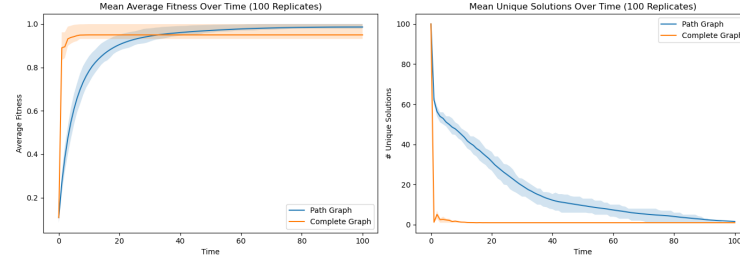
3. Question 3.

- (a)
- i. They describe exploration as the "development of novel solutions" and exploitation as the "utilization of known solutions." They operationalize this in the model by allowing each agent to either "explore" and try a novel solution – represented as a one-bit mutation – or simply copy the solution of their best-performing neighbor.
 - ii. In their model, which is an adaptation of an original made to model gene interactions, they conceptualize N as essentially a binary-encoded space of possible activities, and K as the number of other activities the value of a given activity is dependent on. The NK landscape is the "quality space" of possible solutions; these being comprised of a binary sequence representing an actor's location in "activity space" N . Each activity (index) in that space has a default contribution it provides in isolation, but the effective operating unit of fitness contribution is not activities themselves, but activity *combinations*, so the actual contribution is dependent on K other indices randomly selected at runtime. Flipping a bit in one index will also alter the contributions of the K other indices it is dependent on. The most interesting regimes occur when $0 < K < N - 1$, so the authors generally keep the parameters within this range.
 - iii. Average performance of the actors relative to the global optimum, plotted over time across 1000 simulations for a linear and fully-connected network. The linear network starts worse, but quickly overtakes the fully-connected one as the latter seemingly hits a hard "ceiling" at about 55%.
 - iv. Figure 5 shows the number of unique solutions as a function of time for both networks. Actors in the fully-connected network converge almost instantaneously to the strategy of the best-performing actor. Their performance necessarily plateaus because they're all picking the same strategy. These plots demonstrate the inverse relationship between short-term (exploitation) and long-term (exploration) performance.
- (b) See notebook.
- (c) Verified; see notebook.
- (d) The trajectories mirror what we should expect given the results in the paper. Even though average fitness is not being measured relative to the global optimum, the same pattern emerges: the complete graph instantly converges and quickly loses all diversity, while the path graph converges slower but achieves higher fitness in the long run.

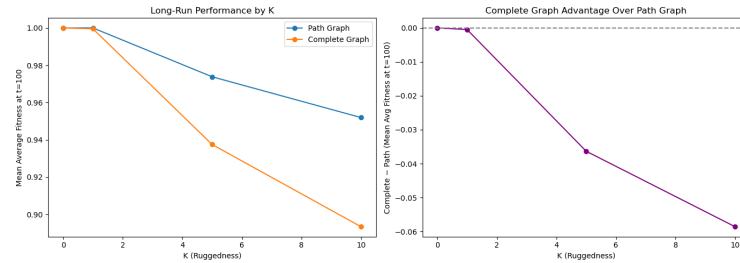


- (e) While not as striking as before (with the seed I specifically chose to make it so), the key result – that the path graph overtakes the complete graph in the

long run – is still clearly present. The complete graph converges rapidly but gets stuck, while the path graph’s slower information diffusion preserves diversity (as shown in the right panel), allowing agents to explore more of the landscape and ultimately reach higher-quality solutions. These results are consistent with the paper’s finding that less connected networks trade short-run with long-run performance.



- (f) K controls the complexity of interactions between bitstrings, so it effectively acts as a slider for the number of local maxima in the landscape (ruggedness). This makes good solutions harder to find for everyone, but especially for groups with fast-convergence strategies. Since the likelihood of randomly happening upon a global optimum becomes smaller and smaller, the complete graph’s ”strategy” of maximally exploiting the best-performing member’s solution gets severely penalized. Increasing K makes exploration increasingly better relative to exploitation.



- (g)
- i. It’s vital to run many trials for the simple reason of statistical defensibility. Running more trials reduces noise, increases confidence, and gets you asymptotically close to the empirical ”mean” or expected value.
 - ii. Controlling for independent variables is a fundamental tenet of the scientific method. The landscape – and any other factors not being tested – need to be held constant because they are just such variables. We are interested in how the network structure alone affects simulation results, so we need to isolate its effect.
 - iii. Scientists often use what might be called ”probe” measurements as a method to test hypotheses concerning the mechanistic underpinnings of earlier measurements. Solution diversity is just such a measurement, in this case pertaining to the observed difference in performance across time.
- (h) I think it’s possible that the paper’s conclusion – that of there being a trade-off between connectivity and innovation – is more an artifact of model design than of reality. The model is set up so that exploration and exploitation are

mutually exclusive, and an inherent preference is assumed for exploitation, with innovation only emerging as a secondary option if the primary strategy of copying doesn't work. With that in mind, I think it would be interesting to see what happens when you allow exploration/exploitation tendencies to vary independently via their own agent-level parameters, potentially even allowing agents to adjust these to optimize for their own individual performance and observing the optimal strategy that emerges in a given network structure as the dependent variable.